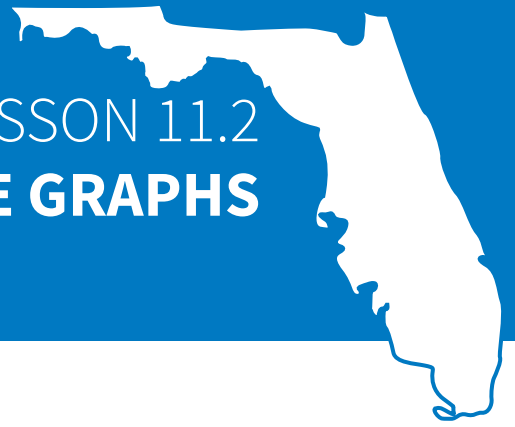


LESSON 11.2

CONSTRUCTING LINE GRAPHS



LESSON INTRODUCTION

MA.8.DP.1.1 Given a set of real-world bivariate numerical data, construct a scatter plot or a line graph as appropriate for the context.

LEARNING TARGETS

- I can construct a line graph given a set of real-world data.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data, and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- How do you construct a line graph?
- How do you use independent and dependent variables when creating a line graph?

LESSON NARRATIVE

This lesson builds on students' previous experiences working with bivariate data and analyzing line graphs. Students will continue determining independent and dependent variables of a data set, as well as selecting a scale for each axis of a graph. Then, students use those details to plot the points to create a line graph.

COMPONENT SUMMARY

11.2.1 WARM-UP (~5 MIN.)

Describe trends

11.2.2 GUIDED INSTRUCTION (15-20 MIN.)

Construct a line graph

11.2.3 ACTIVITY (15-20 MIN.)

Gallery walk

11.2.4 WRAP-UP (~5 MIN)

Create a line graph

RESOURCES

GLOSSARY

Key Terms:

- Trend

Additional Terms: N/A

MATERIALS

- Data sets for the Gallery Walk in 11.2.13 Activity (at least one data set per group)
- Rulers
- Sticky notes

ADDITIONAL PREPARATION

- Have poster-size graph paper available for students who could benefit from visual or fine motor accommodations.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may think that a trend shows exact data, instead of just the general direction of a data set.
- Students may think that an independent variable always causes the event represented by the dependent variable.

THINGS TO CONSIDER

- Discuss “trend” as the general direction of something, much like a ball moves in a general direction when it is kicked or thrown. Remind students of their work in the previous lesson with lines of fit and how they represent a trend.
- Examine several of the graphs in this lesson. Discuss if a cause/effect relationship exists between the independent and dependent variables.

LITERACY AND LANGUAGE ROUTINES

Take Turns

Consider implementing a Take Turns routine when students create the line graphs. This activity provides the opportunity for students to “take turns” examining a data point and then plotting it on a graph. Using a Take Turns routine will give students the opportunity to explain and articulate their process for creating a line graph.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

Encourage students to use the Gallery Walk to analyze the mathematical thinking of others. Ask students to compare the different graphs representing the same data set, and consider if one construction conveys the data more clearly than another. Also, provide opportunities for students to share their mathematical thinking with peers.

DIFFERENTIATE INSTRUCTION

Provide visual supports to help make abstract concepts more concrete. Offer poster-size sheets of graph paper to students who experience visual and/or small motor challenges. Orally discuss how to create a graph before students begin working on it. Allow students to work in flexible groupings based on their needs.

11.2.1 WARM-UP: TRENDS

TEACHER MOVES

It is not the expectation that all students will be able to explain this concept correctly. The meaning of trend will be addressed in the Guided Instruction, so there is no need to correct students here. Rather, teachers should observe what students think the term means, so that the students' current understanding and the given priming questions can be used to build toward understanding of this term in mathematics.

DIFFERENTIATE INSTRUCTION

To make the abstract idea of a trend more concrete, students can create a poster depicting different trends from a specific time period. For example, a "Trends of the '80s" poster could feature photos of neon clothes, scrunchies, big hair, and hair bands.



11.2.1 Warm-Up: Trends

1. The term **trend** is used in many contexts.
 - a. Describe a current fashion trend.
Answers will vary.
Sample response: I think tie-dye is a current fashion trend. It is very popular, and people are tie-dying all kinds of different clothes and accessories.
 - b. Describe what it means for something to be "trending" on Netflix or Twitter.
Answers may vary.
Sample response: If something is trending, it means that it is getting increasingly more views or shares.
 - c. What do you think "trend" means in mathematics in the context of numerical data?
Answers will vary.
Sample response: I think that a trend in mathematics in the context of numerical data means that the data is either increasing or decreasing over time.



11.2.2 Guided Instruction: Constructing Line Graphs

1. As of the 2020-2021 NBA season, Jimmy Butler of the Miami Heat played 10 seasons in the NBA. The table shows his average assists per game for his first 10 seasons. In basketball, a player records an assist when he or she passes the ball to a teammate and that teammate scores a field goal (two-point or three-point shot).

Season (Years)	1 (2011-12)	2 (2012-13)	3 (2013-14)	4 (2014-15)	5 (2015-16)
Assists per Game	0.3	1.4	2.6	3.3	4.8
Season (Years)	6 (2016-17)	7 (2017-18)	8 (2018-19)	9 (2019-20)	10 (2020-21)
Assists per Game	5.5	4.9	4.1	6.0	7.3

- a. Identify the independent and dependent variable in this context.

Independent Variable	<i>The season of Jimmy Butler's career (in years).</i>
Dependent Variable	<i>Assists per game.</i>

On a line graph, the independent variable is represented along the x -axis and the dependent variable is represented along the y -axis.

- b. What range of data values need to be represented along each axis to represent this data set?

Range of x-values	<i>1 to 10</i>
Range of y-values	<i>0.3 to 7.3</i>

11.2.2 GUIDED INSTRUCTION: CONSTRUCTING LINE GRAPHS

TEACHER MOVES

Interpreting the data is not part of this benchmark. However, understanding what the data shows is part of what makes statistics interesting. It is also a factor to consider when determining which type of graph is the best representation based on the data. Discuss with students the relationship between the season number and the actual years of the season. When students are creating their own line graphs, they will need to determine how to represent the data on each axis numerically. In this case, the season numbers are given to them.

ENRICHMENT

Encourage synthesis through questioning:

- Why is the season in years the independent variable? Why is assists per game the dependent variable?
- How does the independent variable affect the dependent variable?
- What scale do you think would be appropriate for the x -axis? Why?
- What scale do you think would be appropriate for the y -axis? Why?

11.2.2 GUIDED INSTRUCTION: CONSTRUCTING LINE GRAPHS (CONTINUED)

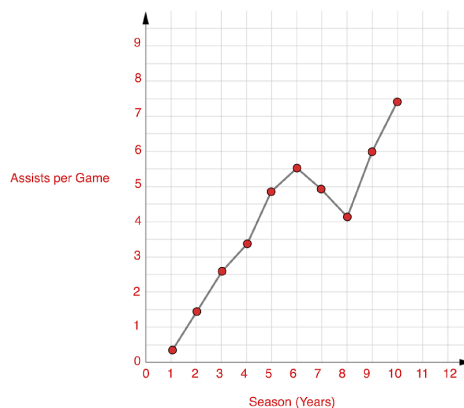
TEACHER MOVES

Discuss the relationship of the independent to the dependent variable in the graph. Work with students as they consider if the season directly affects the assists per game, or if the number of assists per game just happens to occur in that season. Point out that data can be dependent without being directly affected by its corresponding independent data point.

DIFFERENTIATE INSTRUCTION

Provide poster-size sheets of graph paper for students who could benefit from a larger graph. Some students may benefit from orally discussing the steps of completing the graph before they begin working.

- c. Construct a line graph to represent the data by completing the following.
 - Label the axes with the appropriate variables.
 - Identify and label an appropriate scale based on the data for each axis on the grid.
 - Plot a point for each pair of values in the data set.
 - Connect the points in order, using a straightedge.
 - Title the graph.



TEACHER MOVES

When students create their own graphs in the Gallery Walk, they will need to understand how to change given values of time in order to represent them along the x-axis. This can be easily done by labeling the x-axis “Number of (Years, Months, Days, etc.) since _____”

In this case, the years are represented as “Number of Years Since 2011.” Therefore, 0 is 2011, 1 is 2012, 2 is 2013, etc. In Lesson 1, students looked at how to interpret these values from a graph. In this lesson, students will be creating graphs where they have to translate from the data set to the graph.

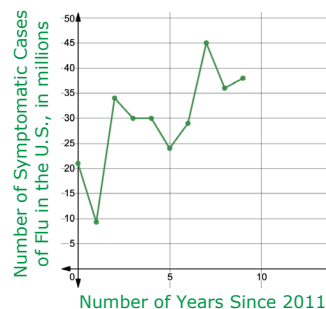
QUESTIONING

In order to challenge early finishers, consider providing students with a more complicated data set to graph.



A **trend** in mathematics describes a change over time. In cases of bivariate data where one of the variables is not time, trends describe how one variable changes in relation to changes in the other.

- d. Describe the trend of the data represented on the line graph.
Answers may vary.
The number of assists per game increases over the first 6 years, decreases the following 2 years, and then increases again. Overall, the number of assists Jimmy Butler has per game increases over the course of his NBA career.
2. A line graph from Lesson 1 is shown. The original data set was given in years such as 2011, 2012, 2013, etc. Discuss with your partner how these values were represented on the x-axis of the graph.

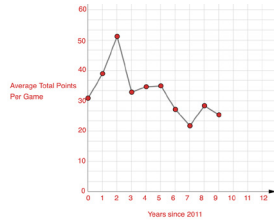




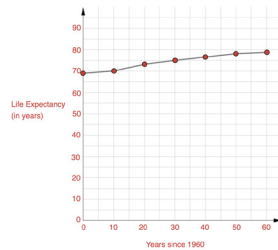
11.2.3 Activity: Gallery Walk

1. Work with your partner or small group to construct a line graph to represent the data set assigned. Be sure to include the following when constructing your graph.
 - Label the axes with the appropriate variables.
 - Identify and label an appropriate scale based on the data for each axis on the grid.
 - Plot a point for each pair of values in the data set.
 - Connect the points in order, using a straightedge.
 - Title the graph.

*Data Set A: FSU
Seminoles Average
Points Per Game*



*Data Set B: US Life
Expectancy*



11.2.3 ACTIVITY: GALLERY WALK

TEACHER MOVES

Students could each be given several sticky notes to leave feedback, at least one Glow comment (positive feedback) and at least one Grow comment (constructive feedback) related to a graph's construction. Examples might include:

Glow examples:

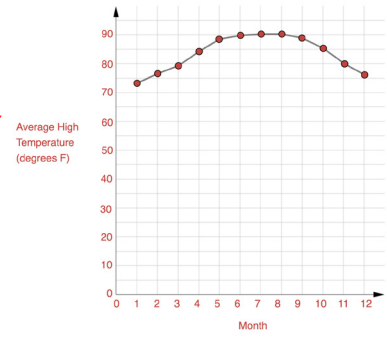
- Titles and labels were clear and appropriate.
- All the points were plotted correctly.

Grow examples:

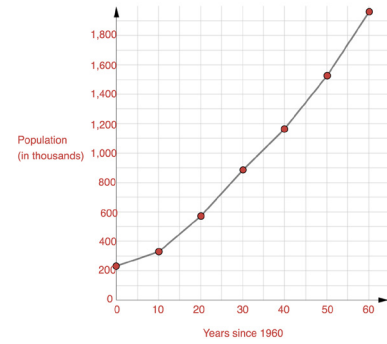
- Consider using a different scale for the ___-axis, perhaps counting by ___ instead of ___.
- It looks like you forgot to label your axes with titles. Be sure to go back to add those so it's clear what your data is showing.

11.2.3 ACTIVITY: GALLERY WALK

*Data Set C:
Average Daily
High
Temperature
in Fort Myers*



*Data Set D:
Population of
Orlando
Metro Area*



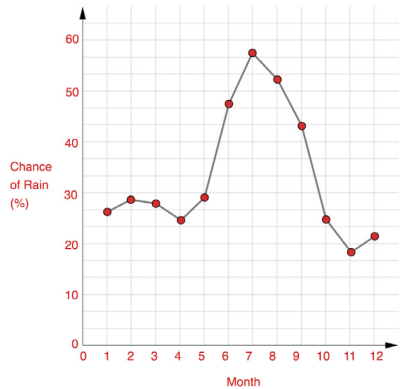


11.2.4 Wrap-Up: Ticket Out the Door

1. Use the data shown to construct a line graph representing the average monthly chance of precipitation, represented as a percentage (%), in Jacksonville, FL.

Month	1	2	3	4	5	6
Chance of Rain (%)	26%	28%	27%	24%	29%	47%
Month	7	8	9	10	11	12
Chance of Rain (%)	57%	52%	43%	25%	18%	21%

Average Daily Chance of Precipitation in Jacksonville, FL.



11.2.4 WRAP-UP: TICKET OUT THE DOOR

DIFFERENTIATE INSTRUCTION

Students should represent the months numerically on the x-axis as discussed in the Guided Instruction and reinforced through the gallery walk. Observational and formative data can be used to determine if more work needs to be done to guide the next lesson.